

Topic 7 Chapters 10 and 11 Worksheet

In the worksheets for this class, you are going to create MySQL statement for each task described below.

You are required to take a screen shot of the SQL statements and the output for each task.

Paste the screenshot of your results under the numbered question.

Save your work and upload the document to the assignment folder by the due date in the Syllabus.

DDL statements

Create three tables using the data below. Consider these points as you set everything up:

- What are the primary keys (one per table)?
 1. judgements: PRIMARY KEY (judgement_id)
 2. food_items: PRIMARY KEY (calorie_id)
 3. consumption: PRIMARY KEY (calorie_id, dates)
- Which are foreign keys?
 1. judgement_id in the food_items referencing judgements (judgement_id): food_items_fk_judgements
 2. calorie_id in the consumption table, referencing food_items (calorie_id): consumption_fk_food_items
- Where would ENUMs be appropriate?

An ENUM data type is useful when a column can contain only a small, fixed set of string values.

 1. judgement: Could use ENUM for the values 'very bad', 'better', and 'excellent'.
 2. unit_measure: Could use ENUM for the values 'piece', 'cup', 'oz', and 'bunch'.
- What other constraints will you add? Apply these constraints *after* you create the table(s).

The following constraints were added using ALTER TABLE:

 1. Foreign Keys: Link food_items to judgements and consumption to food_items.
 2. Composite Primary Key: consumption_pk uses calorie_id and dates as the primary key.
 3. Check Constraint: consumption_chk_units ensures that how_many_units is greater than 0.

Show the SQL code you used to create the tables, modify the tables, and load the tables with data.

- `DROP DATABASE IF EXISTS food_tracking_db;`
- `CREATE DATABASE food_tracking_db;`
- `-- select the database`
- `USE food_tracking_db;`

Create the tables

```

15 • CREATE TABLE judgements
16 (
17     judgement_id INT PRIMARY KEY AUTO_INCREMENT,
18     judgement VARCHAR(10)
19 );
20
21 • CREATE TABLE food_items
22 (
23     calorie_id INT PRIMARY KEY AUTO_INCREMENT,
24     food VARCHAR(50) NOT NULL UNIQUE,
25     unit VARCHAR(10) NOT NULL,
26     unit_measure VARCHAR(20) NOT NULL,
27     calories INT NOT NULL,
28     judgement_id INT NOT NULL
29 );
30
31 • CREATE TABLE consumption
32 (
33     calorie_id INT,
34     how_many_units INT NOT NULL,
35     dates DATE
36 );

```

Modify the tables

```

38 -- Add composite primary key to the consumption table
39 • ALTER TABLE consumption
40 ADD CONSTRAINT consumption_pk
41     PRIMARY KEY (calorie_id, dates);
42
43 -- Add foreign key constraint to food_items referencing judgements
44 • ALTER TABLE food_items
45 ADD CONSTRAINT food_items_fk_judgements
46     FOREIGN KEY (judgement_id) REFERENCES judgements(judgement_id);
47
48 -- Add foreign key constraint to consumption referencing food_items
49 • ALTER TABLE consumption
50 ADD CONSTRAINT consumption_fk_food_items
51     FOREIGN KEY (calorie_id) REFERENCES food_items (calorie_id);
52
53 -- Add check constraint to ensure consumed units are greater than zero
54 • ALTER TABLE consumption
55 ADD CONSTRAINT consumption_chk_units
56     CHECK (how_many_units > 0);

```

Load the tables with data

```

59 • INSERT INTO judgements VALUES
60     (DEFAULT,'very bad'),
61     (DEFAULT,'better'),
62     (DEFAULT,'excellent');
63
64 • INSERT INTO food_items VALUES
65     (DEFAULT,'twizzlers','3','piece',120,1),
66     (DEFAULT,'rice pudding','1/2','cup',302,1),
67     (DEFAULT,'marshmallows','5','piece',124,1),
68     (DEFAULT,'cashew','1','oz',155,2),
69     (DEFAULT,'almonds','1','cup',546,2),
70     (DEFAULT,'broccoli','1','bunch',207,3),
71     (DEFAULT,'green beans','1','cup',34,3);
72
73 • INSERT INTO consumption VALUES
74     (1,2,str_to_date('10/4/2025','%m/%d/%Y')),
75     (5,1,str_to_date('10/4/2025','%m/%d/%Y')),
76     (2,1,str_to_date('9/30/2025','%m/%d/%Y')),
77     (7,1,str_to_date('9/26/2025','%m/%d/%Y'));

```

Then, answer these questions using SQL. Don't just say the answer. What SQL code would you use to answer the questions?

1. What is the judgement (not the id) of each of the food items?

```

9 • SELECT fi.food, j. judgement
10 FROM food_items fi
11 JOIN judgements j ON j.judgement_id = fi.judgement_id;
12

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	food	judgement
▶	twizzlers	very bad
	rice pudding	very bad
	marshmallows	very bad
	cashew	better
	almonds	better
	broccoli	excellent
	green beans	excellent

2. Which food did I consume on October 4? Include the name of the food and the judgement.

```

14 • SELECT fi.food, j.judgement
15 FROM food_items fi
16 JOIN consumption c ON c.calorie_id = fi.calorie_id
17 JOIN judgements j ON j.judgement_id = fi.judgement_id
18 WHERE MONTH(dates) = 10
19 AND DAY(dates) = 4;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
food	judgement			
twizzlers	very bad			
almonds	better			

The tables:

food_items

calorie_id	food	unit	unit_measure	calories	judgement_id
1	twizzlers	3	piece	120	1
2	rice pudding	1/2	cup	302	1
3	marshmallows	5	piece	124	1
4	cashew	1	oz	155	2
5	almonds	1	cup	546	2
6	broccoli	1	bunch	207	3
7	green beans	1	cup	34	3

consumption

calorie_id	how_many_units	dates
1	2	10/4/2025
5	1	10/4/2025
2	1	9/30/2025
7	1	9/26/2025

judgements

judgement_id	judgement
1	Very bad
2	better
3	excellent